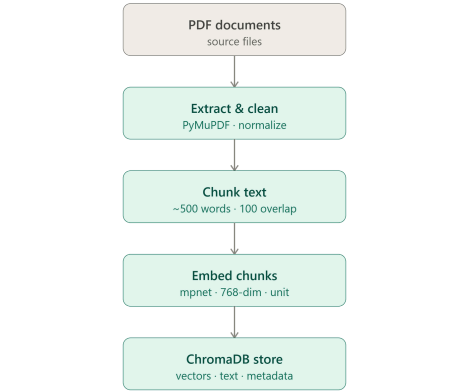
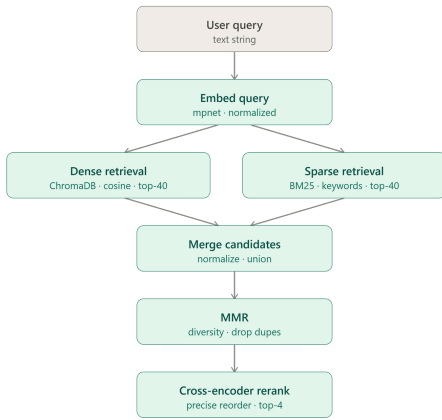


First, ingestion — done once, before any query. This is how your PDFs become searchable vectors sitting in ChromaDB.



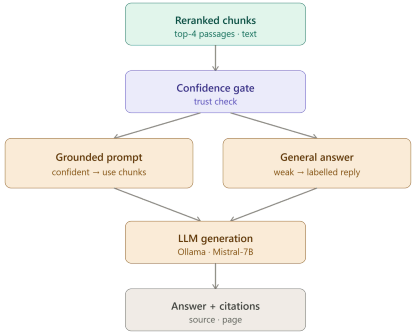
Now the retrieval half of a query. This is the "vector world" — the query gets embedded, searches ChromaDB, and (in parallel) BM25 searches by keyword. The two result sets merge, MMR trims duplicates, and the cross-encoder reorders precisely. What comes out the bottom is a small set of the most relevant chunks, as text.

↓
visualize
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visualize show_widget



Finally, the generation half — the "text world." The reranked chunks hit the confidence gate, which decides whether they're trustworthy. If yes, it builds a grounded prompt from the chunk text; if not, it falls back to a general answer that's clearly labelled as ungrounded. Either way, only text reaches Mistral, and the answer comes back with citations.

↓
visualize
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visualize show_widget



The teal boxes are the vector world, the amber boxes are the text world, and the boundary between them is the single most important architectural fact. Vectors are created for one purpose — searching ChromaDB — and by the time you cross into the amber stages, everything is plain text again. The LLM never sees a vector.

The two parallel retrieval paths converging into the merge is your hybrid search — dense catches meaning, sparse catches exact terms, and neither alone is enough. That fan-in shape is worth being able to draw from memory.

The confidence gate is the one branch in the whole system, and it's what makes your project more than a tutorial RAG. Everything upstream is "find good chunks"; the gate is the moment the system decides whether to trust them before letting the model speak.